FERRIC CARBOXYMALTOSE

Up to 1,000 mg single dose

Hypophosphataemia

CHEMICAL NAME Polynuclear iron(III)-hydroxide carbohydrate (carboxymaltose) complex; 50 mg iron/mL

CLASS Parenteral iron (high-dose, rapid infusion)

TRADE NAMES Ferinject · Injectafer · Iroprem

THERAPEUTIC USES / INDICATIONS

- Iron deficiency when oral iron is ineffective, not tolerated or rapid repletion needed (IBD, heavy menstrual bleeding, pre-operative anaemia).
- Iron deficiency in heart failure (improves symptoms, ↓ hospitalisation).
- Pregnancy (2nd/3rd trimester), postpartum anaemia; CKD (non-dialysis).

ADMINISTRATION

- **IV** bolus (up to 1,000 mg over ≥15 min) or infusion diluted in saline only (max 1,000 mg in ≤250 mL).
- Observe ≥30 min after each dose.
- Avoid extravasation.

DOSAGES

- **Single dose:** max 1,000 mg (or 20 mg/kg) per week.
- Total dose by body weight and Hb (e.g. 1,000–2,000 mg total).
- **Heart failure:** 500–1,000 mg, repeat per ferritin/TSAT.

PHARMACOKINETICS

- Taken up by reticuloendothelial cells; iron utilised for erythropoiesis.
- **t½:** 7–12 h.

PHARMACODYNAMICS

- Stable complex allows controlled release of large iron doses → rapid repletion of stores.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, dizziness, nausea.
- Flushing; injection-site reactions.
- Transient rise in liver enzymes.

ADVERSE EFFECTS (SERIOUS)

- Hypophosphataemia (FGF23-mediated) — can be severe, osteomalacia with repeated dosing.
- Hypersensitivity reactions.
- Skin staining with extravasation.

CONTRAINDICATIONS & CAUTIONS

- Anaemia not due to iron deficiency; iron overload.
- Hypersensitivity to IV iron.
- First trimester of pregnancy.

KEY INTERACTIONS

- **Oral iron:** ↓ absorption; stop during IV therapy.

NURSING MANAGEMENT

- Monitor vital signs and for reactions during and 30 min after.
- Check phosphate in repeated dosing or symptoms (fatigue, bone pain, muscle weakness).
- Recheck Hb, ferritin, TSAT after 4 weeks.

PATIENT EDUCATION

- Report rash, breathing difficulty, dizziness during or after the infusion.
- Report bone pain, muscle weakness or severe fatigue (low phosphate).
- Stop iron tablets unless told otherwise.

KEY POINTS

- High-dose IV iron; up to 1,000 mg per infusion.
- Hypophosphataemia.
- Iron deficiency in heart failure.
- Observe 30 min for hypersensitivity.